

Artificial Intelligence

AI2002

Rushda Muneer

Spring 2026

GENETIC ALGORITHM - Genetic Operators:

Mutation

- Mutation is another important genetic operator
- Mutation takes a **single candidate and randomly changes some aspect (gene) of it**
- For example, **mutation may randomly select a bit** in the pattern and change it, switching a 1 to a 0 or to # (don't care)

GENETIC ALGORITHM - Genetic Operators:

Mutation

- Mutation is important in that **the initial population may exclude an essential component** of a solution
- For example, if no member of the initial population has a 1 in the first position, then crossovers cannot produce a child that could become a solution

GENETIC ALGORITHM - Genetic Operators:

Mutation

- Each gene of each offspring is mutated with a given **mutation rate** p_μ (say 0.01)
- It is hence possible that no gene may be mutated for many generations. On the other hand more than one gene may be mutated in the same generation (or even in the same chromosome)

GENETIC ALGORITHM - Genetic Operators:

Mutation

- For real valued genes, the value is selected randomly from the alleles
- If the rate is too low, new traits will appear too slowly in the population. If the rate is too high, each generation will be unrelated to the previous generation

GENETIC ALGORITHM - Genetic Operators: Crossover (Important Similarities)

- Suppose we have these chromosomes and their fitness values

A)	01101	169
B)	11000	576
C)	01000	64
D)	10011	361

- If we analyze **the strings which have high fitness value, we have observe that they both have a 1 in the 1st column, whereas the low fitness strings have a 0.** In the second column we have a 1 in B and 0 in D, and 1 in the low fitness strings. We wonder that if **1 in the 2nd column is associated with low fitness, so we may improve the score of B by putting a 0 in place of 1 in the 2nd column**

GENETIC ALGORITHM – Selection Algorithm

- It is used to select parents from the current population to be mated with one another to create children
- The selection is primarily based on the fitness. The better the fitness of a chromosome, the greater its chance of being selected to be a parent
- The rate at which a selection algorithm selects individuals with above average fitness is selective pressure
- If there is not enough selective pressure, the population will fail to converge upon a solution. If there's too much, the population may not have enough diversity and converge prematurely.

GENETIC ALGORITHM – Selection Algorithm

- Four types
 - Random Selection
 - Proportionate Selection
 - Linear Rank Selection
 - Tournament Selection

GENETIC ALGORITHM – Selection Algorithm

- Random Selection
- Individuals are selected randomly with no reference to fitness at all
- All individuals, good or bad, have an equal chance of being selected

GENETIC ALGORITHM – Selection Algorithm

- Proportionate Selection
- Chromosomes are selected based on their fitness relative to the fitness of all other chromosomes
- For this all the fitness are added to form a sum S and each chromosome is assigned a relative fitness (which is its fitness divided by the total fitness S)
- A process similar to spinning a roulette wheel is adopted to choose a parent; the better a chromosome's relative fitness, the higher its chances of selection

GENETIC ALGORITHM – Selection Algorithm

- The selection of only the most fittest chromosomes may result in the loss of a correct gene value which may be present in a less fit member (and the only chance of getting it back is by mutation)
- One way to overcome this risk is to assign probability of selection to each chromosome based on its fitness
- In this way even the less fit members have some chance of surviving into the next generation

GENETIC ALGORITHM – Selection Algorithm

- The probability of selection of a chromosome 'i' may be calculated as
- $P_i = \text{fitness}_i / \sum_j \text{fitness}_j$
- Example

Chromosome	Fitness	Selection Probability
• 1	7	7/14
• 2	4	4/14
• 3	2	2/14
• 4	1	1/14

GENETIC ALGORITHM – Selection Algorithm

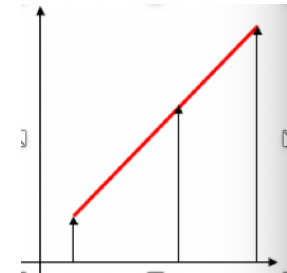
- Advantage
- Selective pressure varies with the distribution of fitness with a population. If there is a lot of fitness difference between the more fit and less fit chromosomes, then the selective pressure will be higher
- Disadvantage
- As the population converges upon a solution, the selective pressure decreases, which may hinder the GA to find better solutions

GENETIC ALGORITHM – Selection Algorithm

- Linear Rank Selection
- Even after selecting chromosomes by their probabilities there is a risk that a chromosome may have a vital gene gone but its fitness may be zero and hence its selection probability will be zero
- The linear rank method assigns a small non-zero probability to even those chromosome which have a zero fitness

GENETIC ALGORITHM – Selection Algorithm

- Linear Rank Selection
- The population is sorted from best to worst according to the fitness
- Each chromosome is then assigned a new fitness based on a linear ranking function
- New fitness = $(P-r) + 1$
- Where P = population size, r = fitness rank of the chromosome
- If $P = 11$, then a chromosome of rank 1 will have a New fitness of $10+1 = 11$ and chromosome of rank 6 will have 6



GENETIC ALGORITHM – Selection Algorithm

- Linear Rank Selection
- A user adjusted slope can also be incorporated
- $\text{New Fitness} = (P-r) (\text{max} - \text{min}) / (P-1) + \text{min}$
- Where max and min are set by the user to determine the slope $(\text{max} - \text{min}) / (P-1)$ of the function
- Let $P = 11$, $\text{max} = 8$ and $\text{min} = 3$,
- Then a chromosome of rank 1 will have a New Fitness of $10 * 5 / 10 + 3 = 8$ and a chromosome of rank 6 will have $5 * 5 / 10 + 3 = 5.5$

GENETIC ALGORITHM – Selection Algorithm

- Linear Rank Selection
- Once the new fitness is assigned, parents are selected by the same roulette wheel procedure used in proportionate selection
- Advantage: selective pressure, once determined by the user remains constant
- Disadvantage:
- Population must be sorted
- Chromosomes with the same fitness will not have the same probability of being selected

GENETIC ALGORITHM – Selection Algorithm

- Another way of Linear Rank Selection is:
- The highest ranking chromosome has probability p of selection (e.g. 0.667)
- If the 1st chromosome fails to get selected, then it is taken out of competition and the second ranking chromosome gets the chance of selection with a probability p among the remaining chromosomes (hence its overall probability is $0.333 \times 0.667 = 0.22$)
- This process is continued for all chromosomes until one gets selected

GENETIC ALGORITHM – Selection Algorithm

- Example

• Chromosome	Fitness	Proportionate Section	Rank	Linear Rank
• 1	7	$7/14 = 0.5$	1	0.667
• 2	4	$4/14 = 0.28$	2	$0.333 * 0.667 = 0.222$
• 3	2	$2/14$	3	$0.333 * 0.333 * 0.667 = 0.074$
• 4	1	$1/14$	4	$0.333 * 0.333 * 0.333 * 0.667 = 0.025$
• 5	0	$0/14 = 0$	5	$1 - 0.667 - 0.222 - 0.074 - 0.025 = 0.012$

Sort n individuals by quality/fitness

Let $p=0.667$ be the probability of selecting i^{th} candidate given that $i-1$ candidates have not been selected

Except the final candidate which is selected if none of the previous candidate has been selected

Select a candidate using the computed probabilities

GENETIC ALGORITHM – Selection Algorithm

- Tournament selection
- One parent is selected by comparing a subset b of the available chromosomes, and selecting the fittest; a second parent may be selected by repeating the process
- The selection pressure increases as b increases
- Value of $b=2$ is most commonly used

GENECTIC ALGORITHM – Survival of the most diverse

- If a chromosome is close to the best chromosome and has a higher score than a chromosome that is far away (diverse) but has a slightly lower score, then the far away chromosome will be wiped out
- This results in uniformity, even in large population

9	1	2	3	4	5	4	3	2	1
8	2	0	0	0	0	0	0	0	2
7	3	0	0	0	0	0	0	0	3
6	4	0	0	7	8	7	0	0	4
5	5	0	0	8	9	8	0	0	5
4	4	0	0	7	8	7	0	0	4
3	3	0	0	0	0	0	0	0	3
2	2	0	0	0	0	0	0	0	2
1	1	2	3	4	5	4	3	2	1
	1	2	3	4	5	6	7	8	9

GENECTIC ALGORITHM – Survival of the most diverse

- Diversity of a chromosome from the already selected chromosomes can be calculated as
- Take the distance of the chromosome from a selected chromosome
- Square the distance (to remove negative distance)
- Take the sum of all squared distances
- The larger this sum the more diverse is the chromosome
- Diversity = $\sum_j \text{distances}_j^2$
- One way to incorporate diversity is to rank the chromosomes by the sum of quality rank and diversity rank

GENECTIC ALGORITHM – Survival of the most diverse

- Hence diversity (or being different) can be incorporated into the selection procedure and a chromosome which is almost as good as another chromosome but is far away from the already selected chromosomes has a better chance of survival

GENECTIC ALGORITHM – Generation Gap

- It is a real number between zero and one the represents the fraction of the current population that gets replaced by the offspring
- If it is 1.0, all the current chromosomes are replaced. This implies that the population P we would have to create P children
- For a generation gap value between 0.0 and 1.0, it is necessary to create that many children, e.g. for a gap of 0.5 we would need to create $0.5 * P$ children

GENECTIC ALGORITHM – Generation Gap

- Another issue is how to determine which individuals in the current population die. The most common strategy is to replace the worst individuals of a population
- We may have a generation gap which is not fixed. In this method the fittest P chromosomes will be selected from the set of current population plus new children, and will form the new generation

GENECTIC ALGORITHM – Amount of Elitism used

- A real value between 0 and 1, which represents the fraction of the best individual of a population that will not get selected to die
- Example $P = 20$ and the elitism = 0.1, then 2 best individuals of current population do not get replaced
- It is only used if generation gap is not fixed
- If generation gap of 0.9 is used that means 18 old chromosomes will get replaced
- If elitism of 0.1 is used that means that any number of old chromosomes between 0 and 18 may get replaced

GENECTIC ALGORITHM – Number of Duplicates Allowed

- Chromosomes that represent the same candidate solution
- If they are allowed then that chromosome has higher probability of producing an offspring, and may probably create many offspring
- Eliminating duplicates increases the efficiency of the genetic search and reduces the danger of premature convergence
- Eliminating duplicates means that if an offspring is created which is a duplicate of a chromosome of the current population, we terminate it immediately and create a new one. It increases the processing time in large populations

GENECTIC ALGORITHM - Question

- In genetic algorithm, a generation has 50 chromosomes, out of which 5 fittest ones are placed in the next generation without any competition from the offspring. The rest of 45 chromosomes of the next generation are chosen from the pool of parents and offspring. Suppose 30 chromosomes of the offspring make it to the new generation. What is the generation gap and what is the elitism value? Duplicates are not allowed
- Answer:
 - Elitism = 0.1
 - Generation Gap = 0.6

GENETIC ALGORITHM - Example

Use GA to learn the concept Yes Reaction from the Food Allergy problem's data

Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
2	Lobdell	lunch	Friday	expensive	no
3	Sam's	lunch	Saturday	cheap	yes
4	Sarah's	breakfast	Sunday	cheap	no
5	Sam's	breakfast	Sunday	expensive	no

GENETIC ALGORITHM - Chromosomes

Encoding

- A potential model of the data can be represented as a chromosome with the genetic representation:

Gene # 1	Gene # 2	Gene # 3	Gene # 4
Restaurant	Meal	Day	Cost

- The alleles of genes are:
- Restaurant gene: Sam, Lobdell, Sarah, X
- Meal gene: breakfast, lunch, X
- Day gene: Friday, Saturday, Sunday, X
- Cost gene: cheap, expensive, X

GENETIC ALGORITHM

- Population Size
- Let the population size be 3 chromosomes/generation, because the instances are very few
- Fitness Function
- The fitness function can be the number of training samples correctly classified by a chromosome (model)

GENETIC ALGORITHM

- Genetic Operators
 - Use the crossover operator, with randomly chosen single point crossover
 - Use mutation on one gene after every two iterations
- Selection Algorithm
 - Proportionate selection

GENETIC ALGORITHM

- Generation Gap
 - Variable, discard the 2 least fit chromosomes
- Elitism
 - No elitism; discard the 2 least fit chromosomes
- Duplicates
 - Duplicates are not allowed
- Criteria for Termination
 - A chromosome correctly classifies all training samples

GENETIC ALGORITHM

1st Iteration

Suppose the following three chromosomes are created randomly:

Sam	X	Sunday	cheap
X	breakfast	Sunday	X
Sam	breakfast	X	X

GENETIC ALGORITHM

Sam	X	Sunday	cheap
-----	---	--------	-------

X	breakfast	Sunday	X
---	-----------	--------	---

Sam	breakfast	X	X
-----	-----------	---	---

The fitness of 1st chromosome is 3

The fitness of 2nd chromosome is 1

The fitness of 3rd chromosome is 3

Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
2	Lobdell	lunch	Friday	expensive	no
3	Sam's	lunch	Saturday	cheap	yes
4	Sarah's	breakfast	Sunday	cheap	no
5	Sam's	breakfast	Sunday	expensive	no

GENETIC ALGORITHM

1st Iteration

Suppose these two chromosomes are selected for crossover:

Sam	X	Sunday	cheap
Sam	breakfast	X	X

Suppose the crossover is after 2nd gene, and we get

Sam	X	X	X
Sam	breakfast	Sunday	cheap

GENETIC ALGORITHM

1st Iteration

Sam	X	X	X
-----	---	---	---

Sam	breakfast	Sunday	cheap
-----	-----------	--------	-------

The fitness of 1st chromosome is: 4
and that of 2nd is: 3

Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
2	Lobdell	lunch	Friday	expensive	no
3	Sam's	lunch	Saturday	cheap	yes
4	Sarah's	breakfast	Sunday	cheap	no
5	Sam's	breakfast	Sunday	expensive	no

We reject the chromosome of fitness 1 and 3 and the rest of the three chromosomes (two old and one new) will go into the 2nd iteration

Must Read

- Chapter 25 Winston